MIL-C-28741A(EC)
27 June 1972
SUPERSEDING
MIL-C-28741(EC)
8 April 1970
(See 6.3)

MILITARY SPECIFICATION

CONTROL MONITOR,

C-8430/UPX

1. SCOPE

1.1 This specification covers the manufacture and test of the C-8430/UPX Control Monitor, to be installed aboard naval vessels.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-L-7806 - Light, Panel, Plastic Plate Lighting
MIL-S-901 - Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-Q-9858 - Quality Program Requirements
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore, General Specification
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment
MIL-STD-781 - Reliability Tests: Exponential Distribution
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipments and Facilities

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposals shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402).

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with the provisions of MIL-E-16400 except as hereinafter specified. Screws and screw thread devices shall be in accordance with Handbook H28.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.3 Simplification and improvement. The requirements of this specification do not preclude simplification and improvement. However, any proposals for simplifications or improvements shall be submitted in writing to the command or agency concerned, and be approved in writing prior to their inclusion (see 6.1.2).

3.4 Reliability.

3.4.1 Reliability prediction. The contractor shall submit a reliability prediction based on stress factors and part populations per MIL-HDBK-217 to the command or agency concerned. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned (see 6.1.2).

3.4.2 Quantitative reliability requirement. The specified mean-time-between-failure, (MTBF) (θ_o as defined in MIL-STD-781), of this equipment shall be 50,000 hours.

3.5 Design requirements.

3.5.1 Environmental conditions. The equipment shall be capable of operation under the conditions of (a) through (d) as specified in MIL-E-16400:

(a) Temperature - Class 2
(b) Shock - Grade A, Class I, lightweight, type A of MIL-S-901
(c) Vibration - Type I of MIL-STD-167. The upper limit for endurable test shall be 50 Hz
(d) Humidity - 95 percent relative humidity.

3.5.2 Synchros. Synchros are not required.

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3.5.3 Controls and indicators. The following controls and indicators shall be provided (see figure 1 for tentative panel lay-out):

Function	Control	Indicator	Color
Panel Lights	x(Screwdriver)		
# Panel Power	x(Circuit breaker)		
Pretrigger Alarm		x	Yellow
Receiver gain	x(Screwdriver)		
Gain Time Control	Long-Short		
HV Overload	Normal-reset	x	Red
Master Control		x	White
Defruiter Control	Defruit-Stdby	xx D-Gr;Stdby-	White
Zeroize	Normal-zeroize	x	Red
Lockout	Norm-Lockout	x	Yellow
Code	A-B		
Verification Bits (2)	Normal-Test		
ISLS Fault		x	Yellow

3.6 Special control provisions.

3.6.1 High voltage overload. The RESET position of the high voltage overload switch shall be spring-loaded to preclude the setting of this switch in the RESET position.

3.6.1.1 Primary power line voltage and frequency. Primary power line voltage and frequency shall be 115 volts, alternating current, (Vac), 60 hertz (Hz), 1ϕ.

3.6.1.2 Overload protection. Overload protection shall be provided utilizing a circuit breaker mounted on the front panel.

3.6.2 Zeroize switch. The zeroize switch shall be provided with a hinged cover to preclude accidental operation. The ZEROIZE position shall be spring-loaded to preclude the setting of this switch in ZEROIZE (open) position. The ZEROIZE switch shall be normally closed to ground.

3.6.2.1 Mode 4 verification bit switches shall be spring-loaded in the TEST position.

3.6.2.2 The mode 4 lockout override switch shall be a lever lock type designed to prevent accidental setting in the OVERRIDE position.

3.6.2.3 Gain Time Control Switch. The gain time control switch shall be a lever lock type to prevent accidental changing of the selection.

3.6.3 Time meter. A time meter is not required.

3.7 Illumination.

3.7.1 Indicator lights. Provision shall be made for dimming all indicator lights from full brilliance to a minimum but readily discernible limit. The

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means for dimming indicator lights shall be approved by the command or agency concerned. The color of individual lights shall be approved by the command or agency concerned.

3.7.1.1 Panel lights. Plastic plate lighting shall be utilized so that panel designations and descriptions shall be readily discernible in a lighted space. Dimming provisions shall be provided from full illumination to zero. Panel lamps and holders shall conform to MIL-L-7806, color: red.

3.7.1.2 Front panel lay-out. The labeling, grouping and lay-out of the front panel shall be as indicated in figure 1, and in accordance with MIL-STD-1472.

3.8 Size. The unit size shall be kept to a minimum consistent with ease in operation and ready identification of indicators and controls. Dimensions shall not exceed 8 inches in height, 10 inches in width and 8 inches in depth.

3.9 Design. The unit shall be designed to provide all indicators and controls on the front panel, with terminal boards in the rear of the interior of the unit. All indicators shall permit renewal of indicator lamps without removal of the front panel. The basic schematic is provided in figure 2. The unit shall be designed to permit mounting on any one of five sides, and mounting information provided as necessary. The design shall be such as to permit electrical entrance from top, bottom, or either side.

3.10 Electrical entrances. Directions only shall be provided for the provision of electrical entrance means, such as stuffing tubes and the like. (Stuffing tubes, and so forth, are not part of this unit.) The type and number of entrance cables and wires shall be provided by the command or agency concerned who shall approve of the above directions.

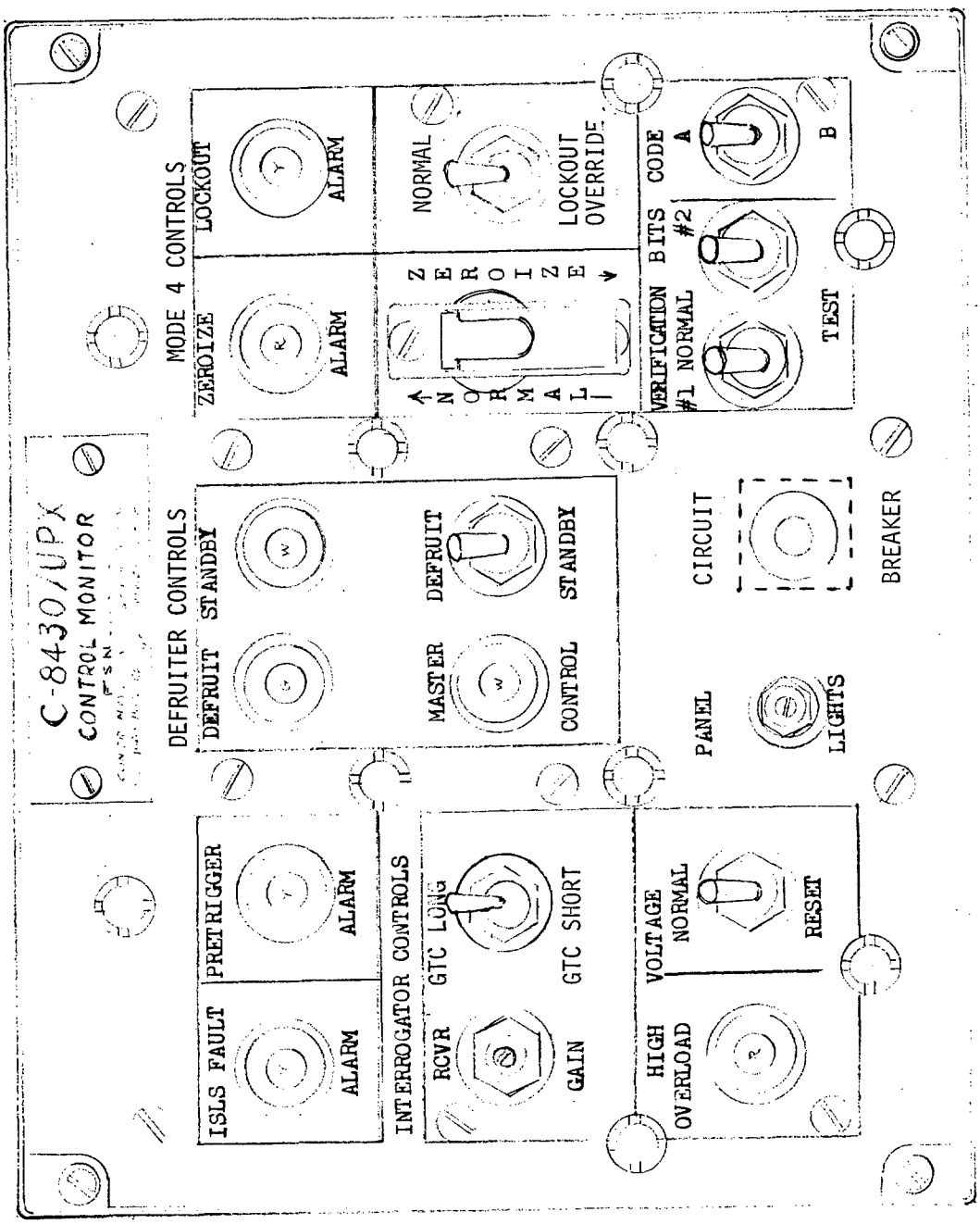
3.11 Interchangeability. The control box shall be completely interchangeable with the prior C-8430/UPX. All parts shall also be directly interchangeable with the corresponding parts of the prior C-8430/UPX. (For the purpose of this requirement C-8430/UPX, Serial C-140C applies).

3.12 Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility. Electromagnetic interference emission, susceptibility equipment and systems self-compatibility is not required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Depth-
Case-6½"
Overall- 8"

8"

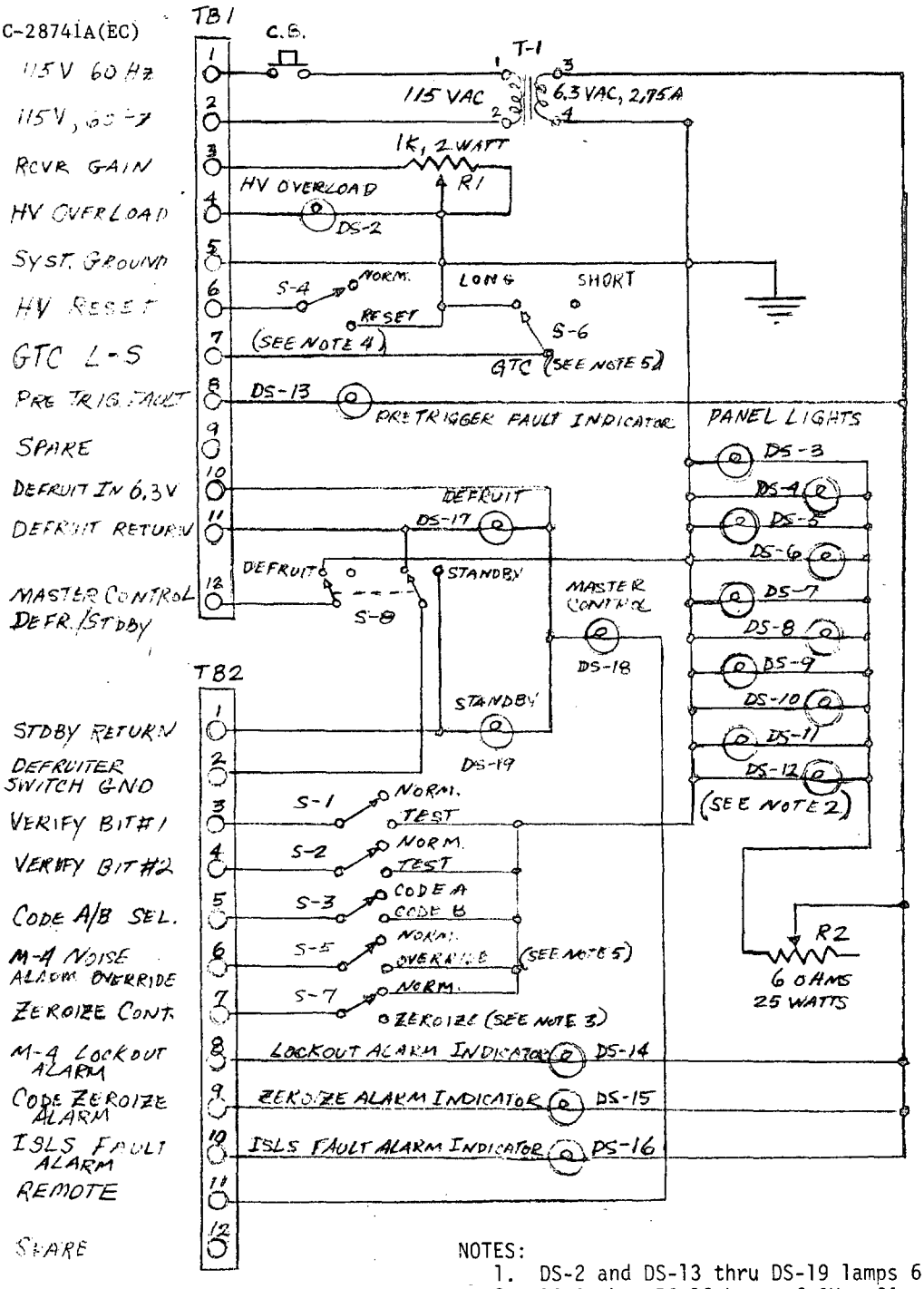
10"

W-WHITE Y-YELLOW G-GREEN R-RED

EC 418

Figure 1. C-8430/UPX Control-Monitor

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EC 419

NOTES:
1. DS-2 and DS-13 thru DS-19 lamps 6.3V, .15A
2. DS-3 thru DS-12 Lamps 6.0V, .2A
3. Spring loaded, momentary closure in Zeroize position and provided with hinged cover
4. Spring loaded, momentary closure in reset position
5. Lever-lock type switch

Figure 2. C-8430/UPX Electrical Schematic

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4.1.1 Quality program requirement. The contractor shall provide and maintain a written quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.2 Quality plan. A quality plan, applicable to the equipments covered by this specification shall be prepared for Government approval (see 6.1.2).

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The contractor shall prepare test procedures for Government approval covering first article and quality conformance inspection (see 6.1.2).

4.1.5 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The methods of examination and testing shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)

(c) Inspection for preparation for delivery (see 4.5)

4.3 First article inspection. First article inspection shall be made on one unit. This inspection shall consist of all examinations and testing necessary to determine compliance with the requirements of this specification and shall include the examination and tests specified under production inspection in MIL-E-16400 (except that the tests for water cooling, enclosure, controls and control circuits, accelerated life, inclination and noise, electromagnetic interference emission and structural integrity and reliability are not required) as approved in accordance with 4.1.4 of this specification.

4.3.1 Demonstration of interchangeability. The first article model of the Control Monitor shall be subjected to examination and test to verify complete physical and functional interchangeability with the Government furnished equipment (GFE) C-8430/UPX. In addition, part interchangeability between the first article model and the corresponding part in the GFE C-8430/UPX shall be verified by examination.

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4.3.2 First article inspection report. A first article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. This inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, tests which detect deviations from design, tests of adjustments, and tests which detect hidden defects of material. Production inspection shall be conducted in accordance with the inspection procedures of MIL-E-16400 and shall consist of the following inspections as approved in accordance with 4.1.4:

Surface examination
Operating test

4.4.2 Production control inspection. This inspection shall conform to table I and electrical performance tests approved in accordance with 4.1.4. This inspection shall be performed on units which have passed production inspection. Sampling shall be in accordance with MIL-STD-105. Special inspection level S-3 shall be used for normal, tightened, and reduced inspection utilizing an AQL of 6.5 percent.

TABLE I. Production control inspection

Examination or test
Weights and dimensions
Supply line voltage and frequency
Heat test

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.2.2 Re-inspection of conforming production control inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed production control may be accepted on contract provided that they are re-subjected to and pass production inspection after repair of all visible damage.

4.5 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation delivery shall be in accordance with MIL-E-17555 (see 6.1.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.1.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Technical manuals including operating instructions, are to be provided with each equipment if required in the contract or order.
(b) Proposals for simplifications or improvements (see 3.3)
(c) Reliability prediction (see 3.4.1)
(d) Quality program plan (see 4.1.2)
(e) Test procedures (see 4.1.4)

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples for those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, AND SO FORTH) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO

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Navy - EC

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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